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ABSTRACT

Accurate prediction of the entrained liquid droplet fraction in an annular two-phase flow regime plays a crucial role for estimating the dryout type critical heat flux to identify the optimized flow characteristics in the thermal systems across different industries. Existing studies have provided different correlations based on the limited experimental data. However, these correlations are applicable to certain operating conditions. Therefore, the present study aims at applying a deep learning method, specifically an artificial neural network (ANN), to enhance the prediction of the entrained liquid droplet fraction. Experimental data from various works on annular flow, covering a wide spectrum of pressure and flow conditions, are utilized for training the ANN model. Eight input variables, viz, superficial gas velocity (J_{SG}), superficial liquid velocity (J_{SL}), gas viscosity (μ_G), liquid viscosity (μ_L), gas density (ρ_G), liquid density (ρ_L), pipe diameter (d) and liquid surface tension (σ_{LV}) are considered as input features. The entrained liquid droplet fraction is the single output feature. The present model employs the Bayesian regularization backpropagation algorithm for training. The present ANN model is compared against the performance of linear regression, decision tree and support vector machine algorithms, and found that the performance of the present Bayesian regularization neural network (BRNN) model is superior within ~7.5% deviation. Further, the BRNN model is coupled with the film mass flow rate model to obtain the axial variation of the liquid film mass flow rate and good agreement is noticed when compared against the experimental data.

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I. INTRODUCTION

The study of annular flow regime is of interest to the design engineers working on heat exchangers, steam generators, refrigeration units, and nuclear power plants. The annular flow regime is characterized by continuously flowing vapor core and the liquid film along the heated wall, as depicted in Fig. 1. The relative velocity between the continuously flowing liquid film and vapor results in interface shear. As a result, the liquid droplets tend to entrain into the vapor core from the liquid film. Increased entrainment of liquid droplets results in the faster depletion of the liquid film. Therefore, accurate prediction of the entrained liquid droplet fraction is instrumental in estimating the pressure drop and the pump requirements in pipelines, designing downstream oil and gas facilities, anticipating dryout conditions in boiling heat transfer, ensuring the efficient cooling of nuclear reactor cores, *etc.*. Higher vapor qualities are associated with increased droplet entrainment due to an increase in the interfacial shear. As the droplet entrainment fraction approaches unity, the liquid film thickness becomes zero leading to the occurrence of critical heat flux (CHF).[1] The location of CHF occurrence can be obtained by developing simple models which are based on the flow topology that monitor the liquid film thickness depletion along the heated wall.[2,3] At CHF, the heated wall temperature shoots up abruptly which may lead to the failure of the component. Further, the annular flow pattern undergoes a transformation into the mist flow regime. Understanding this pattern shift is crucial for plotting the slope of the pressure gradient concerning vapor flow characteristics.

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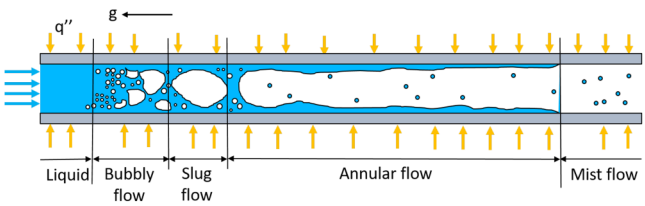

FIG. 1. Schematic of different flow boiling regimes in a vertical tube of a circular cross section.

The phenomenon of liquid droplet entrainment in the context of two-phase annular flow regime is characterized based on the non-dimensional numbers Reynolds number (*Re*) and Weber number (*We*). In the annular flow regime, a dynamic equilibrium is established wherein the liquid droplet deposition and entrainment rates achieve a balance resulting in the dynamic equilibrium of entrainment fraction ($F_{E,\mathrm{Max}}$) of liquid droplets within the gas/vapor core.[4] The entrainment process includes three regions—commencing with the onset of entrainment and concluding with the asymptotic condition. As depicted in Fig. 2, the first region OA is a function of *We* based on the superficial gas velocity. The second region AB is influenced by both *Re* and *We* calculated based on the superficial liquid velocity. The third region BC is solely dependent on *Re*. The overall entrained fraction is quantified by multiplying the local entrainment mass flux by the total area of the gas core, under the assumption that the liquid droplet mass flux is uniformly dispersed within the gas core. In the study by Cioncolini and Thome,[5] the entrainment correlation is described as a liquid atomization process within annular flows. They conceptualized that the gas core in the annular flow regime can be treated as a high velocity confined spray of the gas phase with droplets dispersed in its core at the center. This approach is similar to the aerodynamic interaction and a correlation is established based on this framework.

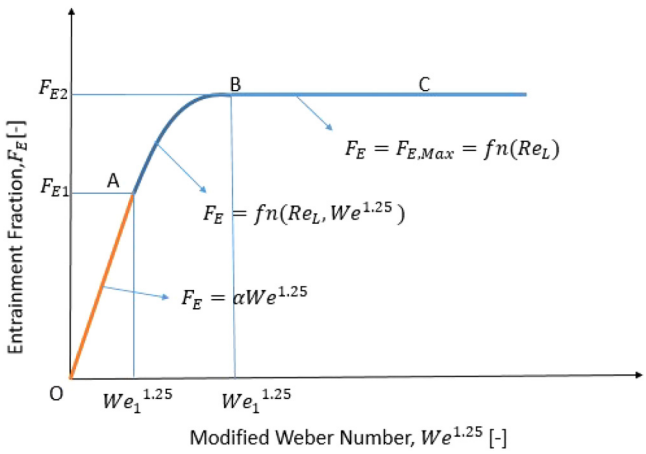

FIG. 2. Schematic of three different phases in the entrainment region as explained by Sawant *et al.*[4]

Modeling of annular flow regime presents challenges such as resolving the thin liquid film flow rate and associated pressure gradient. As an alternative, design engineers prefer to use empirical correlations to estimate the liquid droplet entrainment and deposition rates, liquid film thickness, etc. These correlations are integrated into the thermal hydraulic codes,[6] enabling engineers to identify the critical operating conditions that cause CHF and to prevent potential damage ensuring safe working conditions. However, the accuracy of these numerical model predictions depends on the validity of the closure relations. Although the correlations for the liquid droplet entrainment fraction are widely used and yield accurate predictions, their applicability is limited to a certain operating conditions (see Table I). To overcome the modeling limitations, machine learning (ML) based models are being developed to handle the complex two-phase flow phenomenon.[7–10] To this end, the present study focuses on the application of Bayesian regularization neural network (BRNN) model using the available experimental data in the literature for the accurate prediction of liquid droplet entrainment fraction in an annular flow boiling regime.

II. METHODOLOGY

In the present study, an artificial neural network (ANN) machine learning model based on the Bayesian regularization backpropagation algorithm is employed to ensure that the model does not overfit the data.

1. Bayesian regularization backpropagation algorithm

The Bayesian regularization backpropagation algorithm provides a more robust procedure as compared with other backpropagation algorithms to train neural networks. The robustness of this algorithm is the result of the introduction of Bayesian regularization parameters. For early stopping of the training, the common error function (E_D) is being calculated, which is given as

$$E_D(D|w, M) = \sum_{i=1}^{n} (t_i - \hat{t}_i)^2, \tag{1}$$

where the ANN structure is denoted by M. Here, D can be defined as a dataset with n number of input and output pairs, *i.e.*, $D = \{(u_1, t_1), (u_2, t_2), \ldots.(u_n, t_n)\}$. the size of network training data is n, and t_i is an ith target, w is the weight, and output is denoted by $\hat{t}_i$.

The main reason for the overfitting of the neural network model is the immature convergence. To solve this, the BRNN uses the Bayesian method to regularize the neural network, which uses previous neural network parameter values to optimize the next network values. So for this, the above equation modifies by adding an additional term (E_W) as follows:

$$E_D(D|w, M) = \sum_{i=1}^{n} (t_i - \hat{t}_i)^2 + E_W, \tag{2}$$

where the main function of the additional term of E_W is to perform gradual convergence and give better generalization by

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TABLE I. Correlations for the estimation of liquid droplet entrainment fraction.

S.No.	Model	Correlation	Operating condition
1	Paleev and Filippovich[11]	$1-e=0.985-0.44Log_{10}\left[\frac{\rho^*}{\rho_l}\left(\frac{\mu_l J_g}{\sigma}\right)^2.10^4\right]$, $\rho^*=\rho_g\left[1+\frac{e(1-x)}{x}\right]$	$3000<Re_g<850$ $0.1<C<1.3$, $p\approx 1\ bar$
2	Hewitt and Govan[12]	$e=5.75^*10^{-5}\frac{x}{1-x}\left[\left(G_{lf}-G_{lfc}\right)^2\frac{d\rho_l}{\sigma\rho_g^2}\right]^{0.316}$, $G_{lf}=\frac{\mu_l}{d}exp\left(5.8504+0.4249\frac{\mu_g}{\mu_l}\sqrt{\frac{\rho_l}{\rho_g}}\right)$	$p=1.4-2.4\ bar$ $d=9.53$ mm, 31.8 mm
3	Okawa *et al.*[13]	$e=4.79^*10^{-4}\rho_l\frac{f_i\rho_g j_g^2 h}{\sigma}\left(\frac{\rho_l}{\rho_g}\right)^{0.111}$, $f_i=0.005\left(1+300\frac{h}{d}\right)$	$p=1.4-7.6\ bar$ $T=294-301$ K, $d=5.0$ mm

assigning them unrealistic weights. For minimizing a network, the following gradient-based optimization method is used:

$$F=\beta E_D(D|w,M)+\alpha E_w(w|M), \tag{3}$$

where α and β are the main parameters called hyperparameters for the equation to be optimized and the sum of the square of neural network architecture represented by $E_w(D|w,M)$. This algorithm effectively solves the complex input–output relationship without overfitting the data. The flow chart for the Bayesian regularization artificial neural network (BRNN) training algorithm principle is shown in Fig. 3.

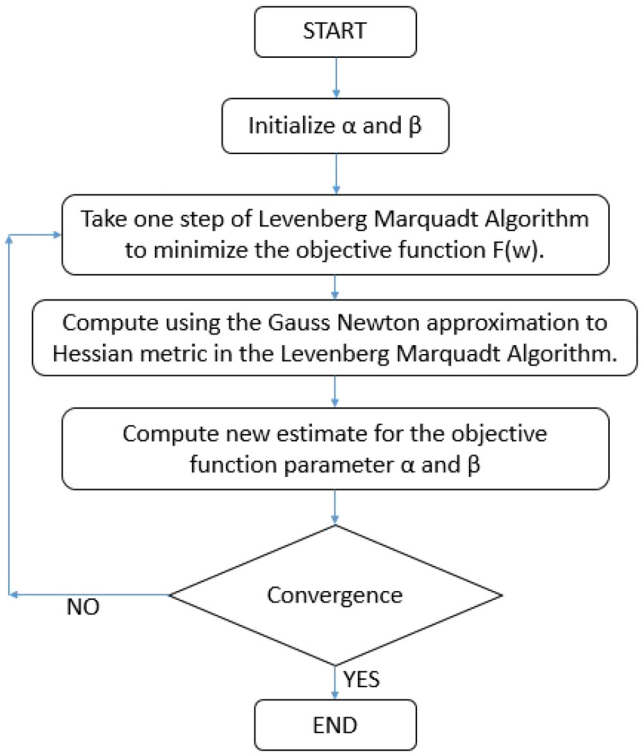

FIG. 3. Flow chart for the Bayesian regularization algorithm.

2. Selection of hyperparameters

The Bayes theorem is being used for the selection of hyperparameter values[14] based on which Eq. (4) is obtained,

$$P(\alpha,\beta\mid D,M)=\frac{P(D\mid\alpha,\beta,M)P(\alpha,\beta\mid M)}{P(D\mid M)}. \tag{4}$$

Here, in Eq. (4), the prior density is assumed to be uniform which is given by $P(\alpha,\beta\mid M)$. In order to maximize the equation, the likelihood function of α and β given by $P(D\mid\alpha,\beta,M)$ is maximized. Using the Bayesian theorem, the density function for the weights is given by Eq. (5),

$$P(\mathbf{w}\mid D,\alpha,\beta,M)=\frac{P(D\mid\mathbf{w},\beta,M)P(\mathbf{w}\mid\alpha,M)}{P(D\mid\alpha,\beta,M)}. \tag{5}$$

For maximizing the likelihood function of α and β, the following Eq. (6) is used:

$$P(D\mid\alpha,\beta,M)=\frac{P(D\mid\mathbf{w},\beta,M)P(\mathbf{w}\mid\alpha,M)}{P(\mathbf{w}\mid D,\alpha,\beta,M)}. \tag{6}$$

The probability densities are given by Eqs. (7) and (8), assuming a Gaussian disturbance for the training data set.

$$P(D\mid\mathbf{w},\beta,M)=\frac{1}{Z_D(\beta)}\exp(-\beta E_D), \tag{7}$$

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$$P(\mathbf{w} \mid \alpha, M) = \frac{1}{Z_W(\alpha)} \exp(-\alpha E_W). \tag{8}$$

The equation for the likelihood function is obtained by substituting the values of Eqs. (7) and (8) in Eq. (6),

$$P(D \mid \alpha, \beta, M) = \frac{\left[\frac{1}{Z_D(\beta)} \exp(-\beta E_D)\right]\left[\frac{1}{Z_W(\alpha)} \exp(-\alpha E_W)\right]}{\frac{1}{Z_F(\alpha, \beta)} \exp(-F(\mathbf{w}))}, \tag{9}$$

$$P(D \mid \alpha, \beta, M) = \frac{Z_F(\alpha, \beta)}{Z_D(\beta) Z_W(\alpha)} \cdot \frac{\exp(-\beta E_D - \alpha E_W)}{\exp(-F(\mathbf{w}))}, \tag{10}$$

$$P(D \mid \alpha, \beta, M) = \frac{Z_F(\alpha, \beta)}{Z_D(\beta) Z_W(\alpha)}. \tag{11}$$

In Eq. (11), the values of $Z_D(\beta)$ and $Z_W(\alpha)$ are determined by Eqs. (12) and (13),

$$Z_D(\beta) = (\pi/\beta)^{n/2}, \tag{12}$$

$$Z_D(\alpha) = (\pi/\alpha)^{N/2}. \tag{13}$$

The value of $Z_F(\alpha, \beta)$ is obtained through the Taylor series expansion and is given by Eq. (14),

$$Z_F \approx (2\pi)^{N/2} \left(\det\left(\left(\mathbf{H}^{\mathrm{MP}}\right)^{-1}\right)\right)^{1/2} \exp\left(-F\left(\mathbf{w}^{\mathrm{MP}}\right)\right), \tag{14}$$

where $\mathbf{H}$ is the Hessian matrix of the objective function,

$$\mathbf{H} = \beta \nabla^2 E_D + \alpha \nabla^2 E_W. \tag{15}$$

To find the optimal values for α and β at the minimum point, the derivative of the log of Eq. (11) is taken and set equal to zero. So the values for α and β is given as „

$$\alpha^{\mathrm{MP}} = \frac{\gamma}{2E_W(\mathbf{w}^{\mathrm{MP}})} \tag{16}$$

$$\beta^{\mathrm{MP}} = \frac{n - \gamma}{2E_D(\mathbf{w}^{\mathrm{MP}})}, \tag{17}$$

where γ is the effective number of parameters used, given by Eq. (18). The total number of parameters in the model is represented by N,

$$\gamma = N - 2\alpha^{\mathrm{MP}} \mathrm{tr}\left(\mathbf{H}^{\mathrm{MP}}\right)^{-1}. \tag{18}$$

With each re-estimation of the objective function parameters, the objective function is changed, shifting the minimum point.[15] The performance function iteratively guides the process toward the next minimum, with parameter updates resulting in increased accuracy. Ultimately, the objective function stabilizes, indicating convergence where negligible changes occur between successive iterations.

3. Artificial neural networks

Artificial neural networks (ANNs) have been developed as computational models inspired by the structure and functioning of the human brain. The neural network model for this study has been empirically decided to have an input layer with 8 input parameters, 1 hidden layer consisting of 12 neurons, and 1 output layer. Each layer of the model is connected to each other with interconnected neurons similar to the human brain. Each neuron connected to the network has a specific weight and bias that influence its role in the training of the network and the output.

During the training process of neural networks, they are provided with a large dataset. The weights and biases of the network are iteratively optimized to minimize the model performance evaluation parameters, viz, R^2 (coefficient of determination) or MSE (mean squared error) or MAPE (mean absolute percentage error). The optimization process adjusts the weights and biases to improve the performance of the model on the training data. The ANN architecture is described in Fig. 4, where the input parameters are denoted by $x_1, x_2, \ldots, x_i$. The neurons which connect these parameters to the network are given weights and bias denoted by W_{ij} and b_j, respectively, and the output combiner for the network is denoted by a_j. The activation function between hidden and output layer is linear and the output for the neuron is denoted by Y_j.

In a mathematical form, the jth neuron is described as

$$a_j = \sum_{i=1}^{8} W_{ij} x_i + b_j. \tag{19}$$

The input values first enter the input layer and then the hidden layer on the basis of different weights and biases to the neurons. The hidden layer consists of twelve neurons, labeled h_j, and the output layer (O_k) has a single neuron, labeled e_k which represents the entrained droplet fraction (e). The activation function

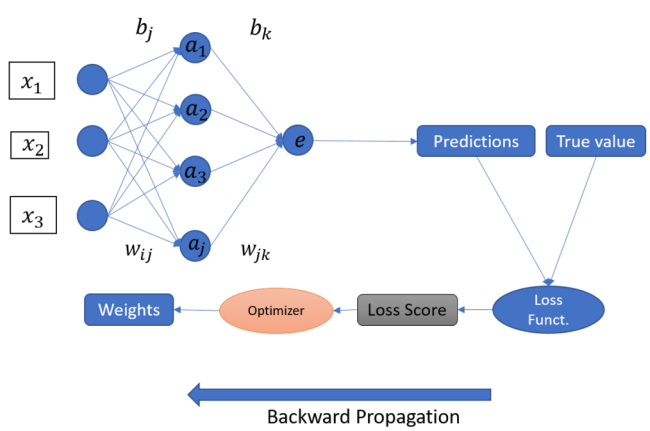

FIG. 4. Schematic of ANN architecture.

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TABLE II. Weights and Bias for the output layer.

W_{jk}		b_i
W_{11}	3.1910	−5.1864
W_{21}	6.331	
W_{31}	−3.9085	
W_{41}	7.8980	
W_{51}	1.171	
W_{61}	3.795	
W_{71}	−4.0596	
W_{81}	−1.202 66	
W_{91}	−7.2029	
W_{10-1}	6.203	
W_{11-1}	3.9622	
W_{12-1}	1.1668	

for the hidden layer is the sigmoid function which is given as

$$Z_j = \frac{2}{1 + e^{-2a_j}} - 1. \tag{20}$$

Then, the output signal of the entrained liquid droplet fraction is calculated as

$$e_k = \sum_{12}^{j=1} W_{jk}{}^{\star} Z_j + b_k. \tag{21}$$

From Eqs. (5) and (6),

$$e_k = \sum_{12}^{j=1} \sum_{1}^{k=1} W_{jk} \left(\frac{2}{1 + e^{-2\left(\sum_{i=1}^{8} W_{ij}{}^{\star} x_i + b_j\right)}} - 1 \right) + b_k. \tag{22}$$

So, the final value of e can be predicted by Eq. (7) where the value of W_{ij} and b_j are weights and bias, respectively, given in Table II. Overall, ANNs offer a flexible and powerful approach for regression tasks, allowing for modeling complex non-linear relationships. By iteratively optimizing the weights and biases, ANNs can learn from data and make accurate predictions.

III. DATASET DESCRIPTION

In the present study, the dataset is obtained from the experimental data available in the literature as presented in Table III. In this study, eight input parameters are used which are superficial gas velocity (J_{SG}), gas viscosity (μ_G), surface tension (σ_{LV}), liquid viscosity (μ_L), gas density (ρ_G), superficial liquid velocity (J_{SL}), gas viscosity (μ_G), tube diameter (d), and liquid density (ρ_L). The entrained liquid droplet fraction is the output parameter. The complete dataset for training of neural networks was divided into two sets: training and testing. The training set, which comprised 90% of the data (1230 × 9), was used to train the neural network by adjusting its weights and biases. This process involved feeding the network with input–output pairs from the training set and updating its parameters to minimize the difference between predicted and actual outputs. The testing set, which accounted for 10% of the data (137 × 9), was used to evaluate the final performance of the trained network. This set was kept separate from the training and validation sets to provide an unbiased assessment of the network's ability to generalize to new or unseen data. By evaluating the network on the testing set, its performance can be assessed in the real world data and make informed decisions about its deployment. In summary, dividing the data into training and testing sets allowed for a systematic approach to developing, fine-tuning, and evaluating the neural network. It ensures that the network learns from the training data and generalized well to new data, leading to a more reliable and accurate model.

IV. MODEL PERFORMANCE

To explain the effect of different input parameters on the predicted value of the neural network, a sensitivity analysis is performed. In the sensitivity analysis, each input parameter is sequentially eliminated, and the R^2 and RMSE values are noted as shown in Fig. 5. It is observed that the tube diameter D has the maximum effect on the predicted value of the BRNN model, and the density of the liquid and surface tension have a lesser impact on the predicted value. However, eliminating the input parameters from the input layer of the neural network reduces the accuracy of the model. The predicted value by the model gives higher values of error than the current model. Therefore, all the input variables are considered important to obtain the best neural network model.

In order to verify the effectiveness of the model, the error in the BRNN output for both the training and testing datasets is observed. As shown in Fig. 6, the black dashed line represents the ideal or exact fit (with 45° slope). Whereas the colored lines (in blue, green and red) show the actual fit by the BRNN model. From the parity plot, it can be concluded that the training, testing, and complete datasets are predicted with very good agreement with the experimental data with R^2, MSE, and RMSE as 0.935 18, 0.0055, and 0.074 16 respectively.

It can be seen that the zero mean Gaussian distribution is obtained for the error which is calculated by taking the difference between the BRNN predicted output and the value fed to the network. The distribution of error of the BRNN model is given in Fig. 7. In this distribution, the output error follows a Gaussian distribution with a mean value close to zero. As it can be observed, Fig. 7 shows 20 bins, i.e., the entire error range is divided into 20 divisions. The leftmost bin or range is 0.5452 and the rightmost bin is 0.2502. The majority of the data points fall within the error range of (− 0.08, 0.08) from the complete dataset. This type of Gaussian distribution of error helps to decide the neural network model prediction performance. As the error approaches the global minima, the present BRNN model can be applicable for a wide range of parameters.

The predictions of the present BRNN model are also compared against the existing correlations that are listed in Table I as shown in Fig. 8. All three correlations are used to predict the entrainment liquid droplet fraction for the complete dataset. It can be noticed that the performance of the present BRNN model is far better than the three correlations. Further, the present model predictions are compared against the conventional ML algorithms

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TABLE III. Experimental data of liquid droplet entrainment in an annular flow regime used for building the model.

S.No.	Author	d (mm)	L/d	System pressure (*bar*)	J_{SL}	J_{SG}	Data points
1	Andreussi *et al.*[16]	24	N/A	1.2	0.37–2.95	5–70	9
2	Wicks *et al.*[17]	28.58	245	1.2	0.13–0.78	22–89	54
3	Cousins *et al.*[18]	32	480	1.2–4.0	0.008–0.01	19–42	40
4	Wurtz[19]	17	450–900	30–90	0.26–3.0	10–67	122
5	Ueda *et al.*[20]	10	245	3.3	0.23–0.68	4–17	27
6	Andreussi[16]	24	208	1.5	0.02–0.44	39–68	33
7	Zabaras *et al.*[21]	50.8	90	1.7	0.006–0.06	9–27	26
8	Fore *et al.*[22]	50.8	69	1.0	0.006–0.06	16–36	65
9	Azzopardi *et al.*[23]	38	118	1.5	0.02–0.1	15–30	28
10	Jepson *et al.*[24]	10.26	292	1.5	0.08–0.14	33–67,44–76	27
11	Lopez de Bertodano *et al.*[25]	10	420	2.8	0.74–0.54	25–126	20
12	Lopez de Bertodano *et al.*[26]	10	350	1.4–6.6	0.1–0.35	4–25	48
13	Feldhaus *et al.*[27]	9.35	241	2.8	0.3–0.65	30–86	14
14	Pan *et al.*[28]	25.4	N/A	1.2	0.04–0.08	32–116	10
15	Azzopardi *et al.*[23]	38	118	1.5	0.02–0.1	15–30	28
16	Schadel *et al.*[29]	25	150	1.2	0.18–1.04	32–116	32
17	Owen[30]	32	600	2.0–4.0	0.02–1	14–100	97
18	Kaji *et al.*[31]	19	300	1.2	0.03–0.65	10–34	32
19	Sawant *et al.*[4]	9.4	210	1.2–4.0	0.05–0.5	16–95	77
20	Al -Yarubi[32]	15	40	1.0	…	…	70
21	Azzopardi *et al.*[23]	125	40	1	0.005–0.02	30–52	19
22	Aliyu *et al.*[33]	101.6	47	1.0–1.4	0.1–1.4	10–29	26
23	Van der Meulen[34]	127	87	2	0.02–0.7	10–17	31
24	Okawa *et al.*[35]	5	320	1.4–7.6	0.1–1.6	20–105	170
25	Adamson *et al.*[36]	14	194–261	70	0.28–1.5	15–20	153
26	Wallis[37]	12.7	N/A	1.0	0.042–0.168	20–45	41
27	Singh *et al.*[38]	12.5	180	69–83	0.18–0.9	8–16	15
28	Langner *et al.*[39]	14	357	11.0	0.01–0.38	2.3–6.8	20
29	Zabaras *et al.*[21]	50.8	69	1.0	0.006–0.06	9–27	26

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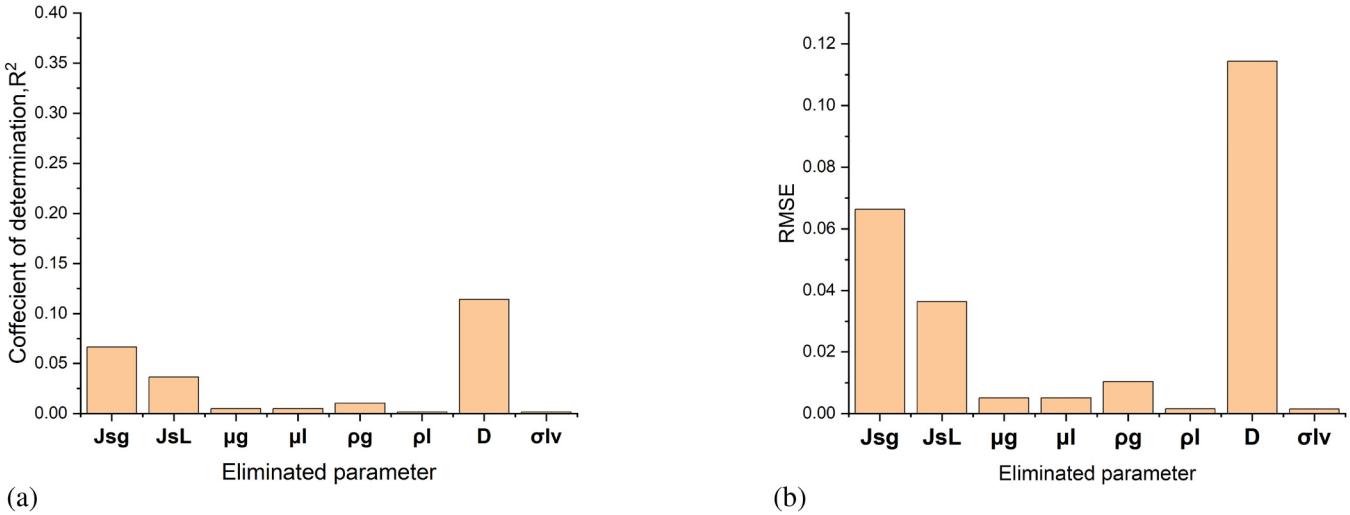

FIG. 5. Sensitive analysis on the basis of (a) R^2 and (b) RMSE.

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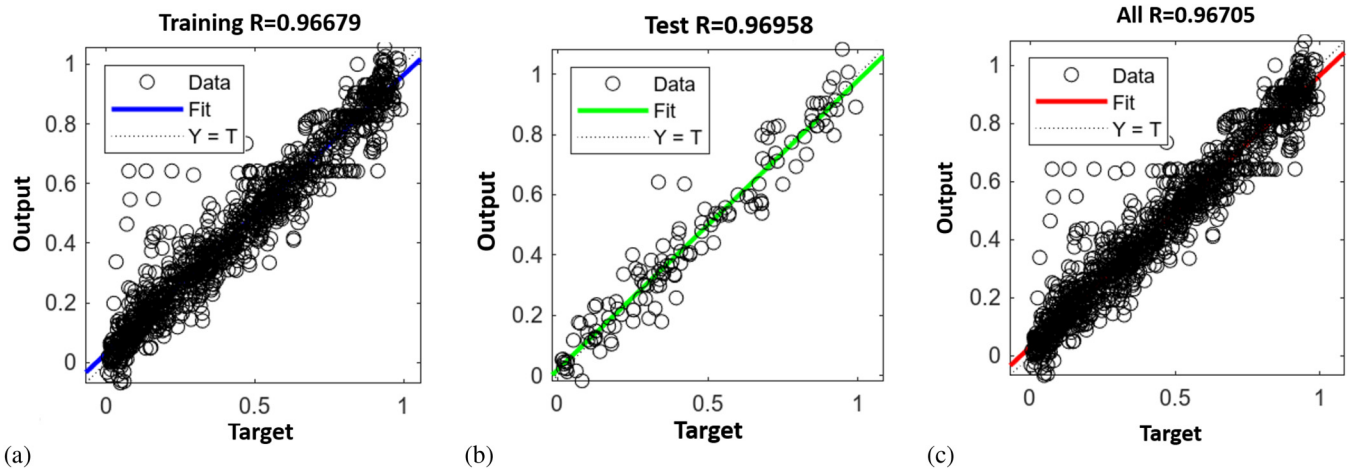

FIG. 6. Performance of the BRNN model for the prediction of liquid droplet entrainment fraction. (a) Training data, (b) testing data, and (c) complete dataset.

such as linear regression, decision tree, and support vector machine using the same dataset. As shown in Table IV, better agreement is obtained by the present BRNN model compared to the conventional ML models.

V. LIQUID FILM MASS FLOW RATE

The mass flow rate of continuously flowing liquid film in the annular flow regime depends on the (i) liquid film evaporation rate, (ii) liquid droplet entrainment into the vapour core and (iii) liquid droplet deposition onto the liquid film. Among these three parameters, the first two aids the liquid film depletion and the latter one aids the liquid film thickness. Along the heated surface, the liquid film depletes in the annular flow boiling regime, and the liquid film thickness eventually approaches zero. The location at which the liquid film thickness becomes zero is considered as the CHF location. At this location, the wall temperature shoots-up due to the poor heat transfer characteristics of the vapour phase. Therefore, the prediction of liquid droplet entrainment rate and the variation of liquid film thickness dictates the CHF ocurrence in thermal systems and energy applications. For calculating the axial variation of the liquid film thickness from the onset of liquid film

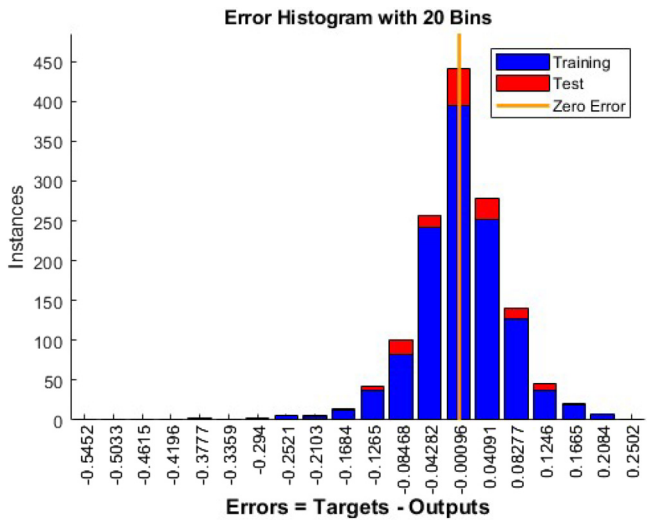

FIG. 7. Error histogram of the present ANN model [8-12-1].

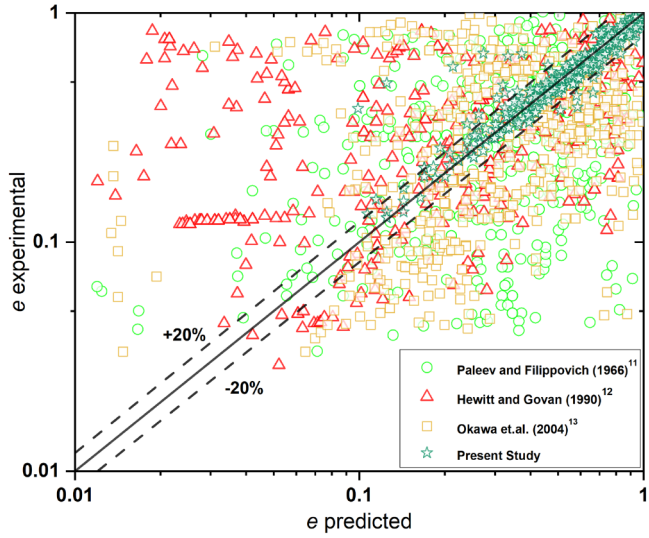

FIG. 8. Comparison of the present ANN model predictions with different correlations as listed in Table I for the liquid droplet entrainment fraction in the annular flow boiling regime.

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TABLE IV. Comparison of the ANN model with different ML models.

Method	RMSE	MSE	R^2
Linear regression	0.2526	0.0638	0.2699
Decision tree	0.1792	0.0289	0.6713
Support vector machine	0.2477	0.0698	0.283
Present BRNN model	0.074 16	0.0055	0.9351

to the CHF location, the following mass conservation equation is solved:

$$\nabla_s.\left(\rho_f h u_f\right) = S_{m,dep} - S_{m,ent} - S_{m,vap}. \tag{23}$$

Here, in Eq. (8), $S_{m,dep}$ represents the mass source term due to deposition of liquid droplets, $S_{m,ent}$ represents the entrainment mass source term, and $S_{m,vap}$ represents the mass source terms due to evaporation. In order to calculate the mass source term due to entrainment, different correlations as listed in Table I are used along with the current BRNN model correlation. The calculated film mass flow rate using different correlations is compared with the experimental data,[36] and very good agreement is achieved as shown in Fig. 9. The calculated liquid droplet entrainment and deposition rates, and the liquid film evaporation rate along the axial length is presented in Fig. 10(a). It was observed that the liquid droplet deposition rate decreases, whereas the liquid droplet entrainment and film evaporation rates increase along the axial length. As a result, the liquid film thickness continuously reduces and the location at which the liquid film thickness completely dries out ($h \approx 0$ mm) is identified as the dryout type CHF location.

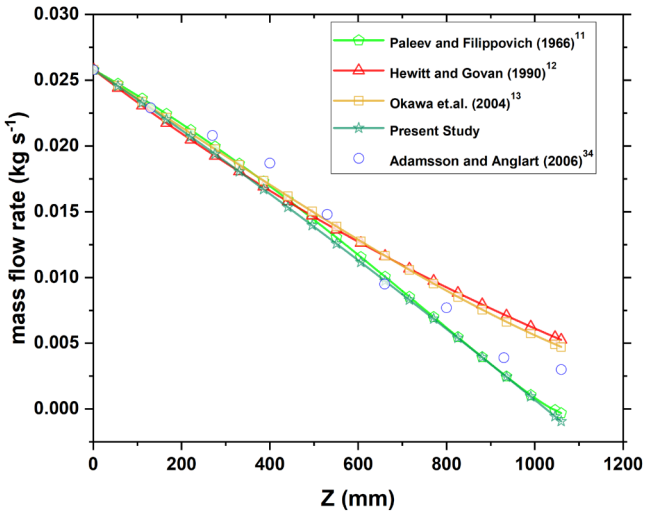

FIG. 9. Comparison of axial variation of liquid film mass flow rate in an annular flow boiling regime against the experimental data[36] using the present model against different correlations listed in Table I.

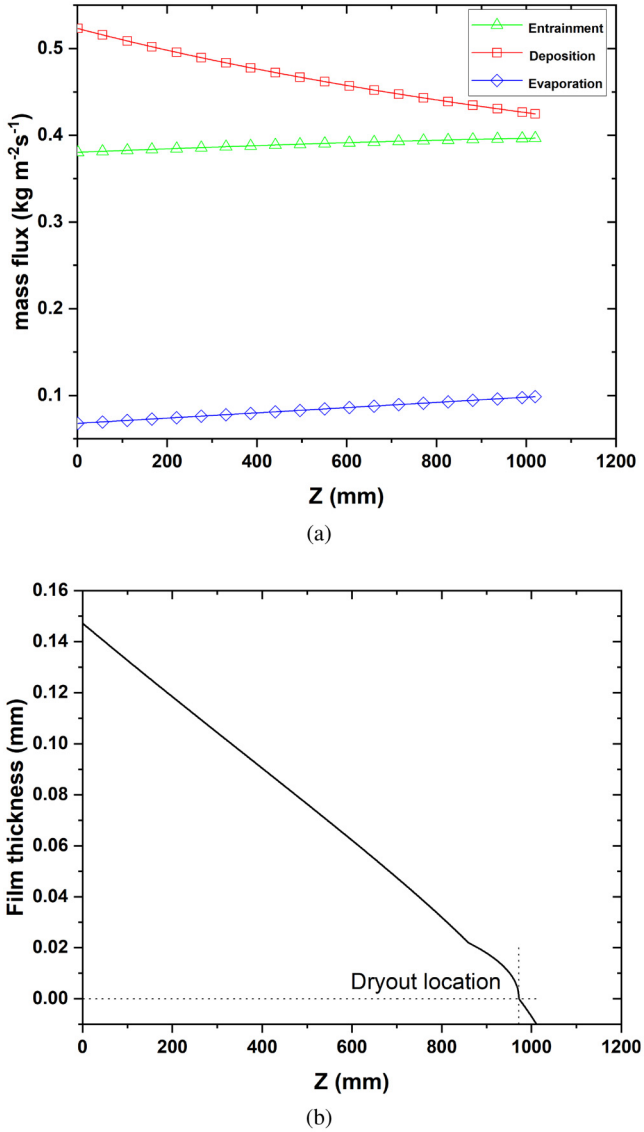

FIG. 10. Axial variation of (a) liquid film evaporation, liquid droplet entrainment, and deposition rates and (b) liquid film thickness for the applied wall heat flux $= 1.032\,\mathrm{MW\,m^{-2}}$.

The axial variation of the liquid film thickness for heat flux, $q = 1.032\,\mathrm{MW\,m^{-2}}$ is shown in Fig. 10(b), wherein the dryout location is identified as $Z_{cr} = 972$ mm.

VI. CONCLUSIONS

In the present study, an artificial neural network based on the Bayesian regularization backpropagation algorithm is developed with eight input parameters and one output parameter of entrained liquid droplet fraction. Different experimental data available in the

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literature are used to train the BRNN model effectively, as shown in Table III. The sensitivity of different input parameters on the output of BRNN model is also compared and found that the diameter of the channel has significant influence on the output variable among the input features. The developed model is compared with the existing correlations[11–13] and found that performance of the current model is better than the existing correlations. The values of R and $RMSE$ for BRNN model are obtained as 0.967 and 0.074 16, respectively. Further, different machine learning techniques such as, linear regression, decision tree, and support vector machine, are compared with the present BRNN model on the basis of RMSE and R^2. It is identified that the present BRNN model performs better than the remaining ML models. The BRNN model for the liquid droplet entrainment is coupled with the film thickness model to obtain the axial variation of liquid film mass flow rate in the annular flow boiling regime. Very good agreement is observed compared with the existing liquid droplet entrainment correlations. Further, the liquid film dryout location is identified as $Z_{cr} = 972$ mm for a heat flux of 1.032 MW m^{-2}.

Although the present model is developed based on the experimental data, there is a need for the improvement of the experimental database. Majority of the experimental studies have focused on small diameter tubes ($d < 100$ mm), indicating a need for more research on the behavior of the annular flow regime in larger dimension systems widely used in industrial settings. Further, it is important to consider the diabatic investigations including a broader selection of coolants such as, refrigerants, liquid metals, *etc.*, to obtain valuable insights to wider applications, viz, heat exchangers, steam generators, nuclear fuel rod bundles, *etc.* To address these limitations, it is very much necessary to conduct non-intrusive optical diagnostic based experiments for the accurate prediction of liquid droplet entrainment which can establish a more robust experimental database for the study of the annular flow boiling regime. Further, to ensure the robustness of the developed correlations, a careful assessment of the existing closure models based on the extensive validation studies is required.[1,40]

NOMENCLATURE

e	Entrained droplet fraction …
h	Liquid film thickness (m)
d	Tube diameter (mm)
p	Pressure (*bar*)
$\mathbf{w}$	Vector of network weights …
G	Mass flux (kg m^{-2} s^{-1})
C	Droplet concentration in gas core …
D	Dataset …
E_W	Sum of squares of network weights …
$\mathbf{H}$	Hessian matrix …
J_{SG}	Superficial gas velocity (m s^{-1})
J_{SL}	Superficial liquid velocity (m s^{-1})
R^2	Coefficient of determination …
Re	Reynolds number …
T	Temperature (K)
We	Weber number …
ρ_G	Gas density (kg m^{-3})
ρ_L	Liquid density (kg m^{-3})
μ_G	Gas viscosity (Pa s)
μ_L	Liquid viscosity (Pa s)
σ	Liquid surface tension (N m^{-1})

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jayesh Vyas: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **Rishika Kohli:** Investigation (supporting); Methodology (supporting); Visualization (supporting); Writing – original draft (supporting). **Shaifu Gupta:** Conceptualization (supporting); Methodology (supporting); Project administration (supporting); Supervision (equal); Writing – review & editing (equal). **Harish Pothukuchi:** Conceptualization (lead); Formal analysis (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Supervision (equal); Validation (supporting); Visualization (supporting); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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